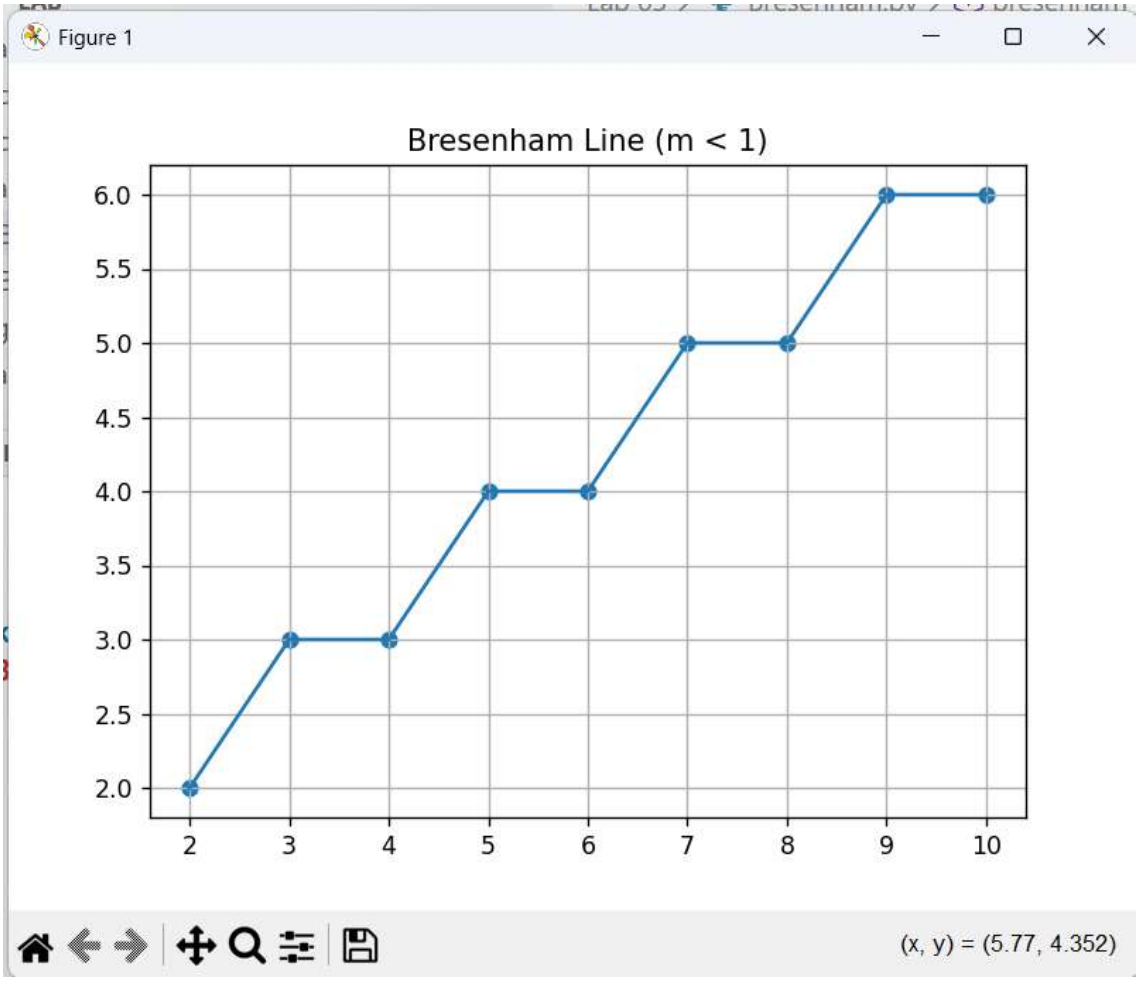
Lab Assignment – 03 || 23103163

A. Develop a program to draw a line using Bresenham's line drawing technique for a slope (m < 1) condition.

```
import matplotlib.pyplot as plt

def bresenham_m_less_than_1(x1, y1, x2, y2):
    points = []

    dx = x2 - x1
    dy = y2 - y1

    p = 2 * dy - dx
    x = x1
    y = y1

    points.append((x, y))

    while x < x2:
        x += 1
        if p < 0:
            p = p + 2 * dy
        else:
            y += 1
            p = p + 2 * dy - 2 * dx

        points.append((x, y))

    return points


# Example
x1, y1 = 2, 2
x2, y2 = 10, 6

points = bresenham_m_less_than_1(x1, y1, x2, y2)

# Plotting
x_vals = [p[0] for p in points]
y_vals = [p[1] for p in points]

plt.scatter(x_vals, y_vals)
plt.plot(x_vals, y_vals)
plt.title("Bresenham Line (m < 1)")
plt.grid()
plt.show()
```



B. To implement Bresenham's line drawing algorithm for drawing straight lines by considering all slope cases.

```
import matplotlib.pyplot as plt

def bresenham_all_cases(x1, y1, x2, y2):
    points = []

    dx = abs(x2 - x1)
    dy = abs(y2 - y1)

    sx = 1 if x2 > x1 else -1
    sy = 1 if y2 > y1 else -1

    if dx > dy:
        p = 2 * dy - dx
        while x1 != x2:
            points.append((x1, y1))
```

```
            x1 += sx
            if p < 0:
                p += 2 * dy
            else:
                y1 += sy
                p += 2 * dy - 2 * dx
    else:
        p = 2 * dx - dy
        while y1 != y2:
            points.append((x1, y1))
            y1 += sy
            if p < 0:
                p += 2 * dx
            else:
                x1 += sx
                p += 2 * dx - 2 * dy

    points.append((x2, y2))
    return points


# Example
x1, y1 = 2, 2
x2, y2 = 6, 12   # try different slopes

points = bresenham_all_cases(x1, y1, x2, y2)

x_vals = [p[0] for p in points]
y_vals = [p[1] for p in points]

plt.scatter(x_vals, y_vals)
plt.plot(x_vals, y_vals)
plt.title("Bresenham Line (All Slopes)")
plt.grid()
plt.show()
```

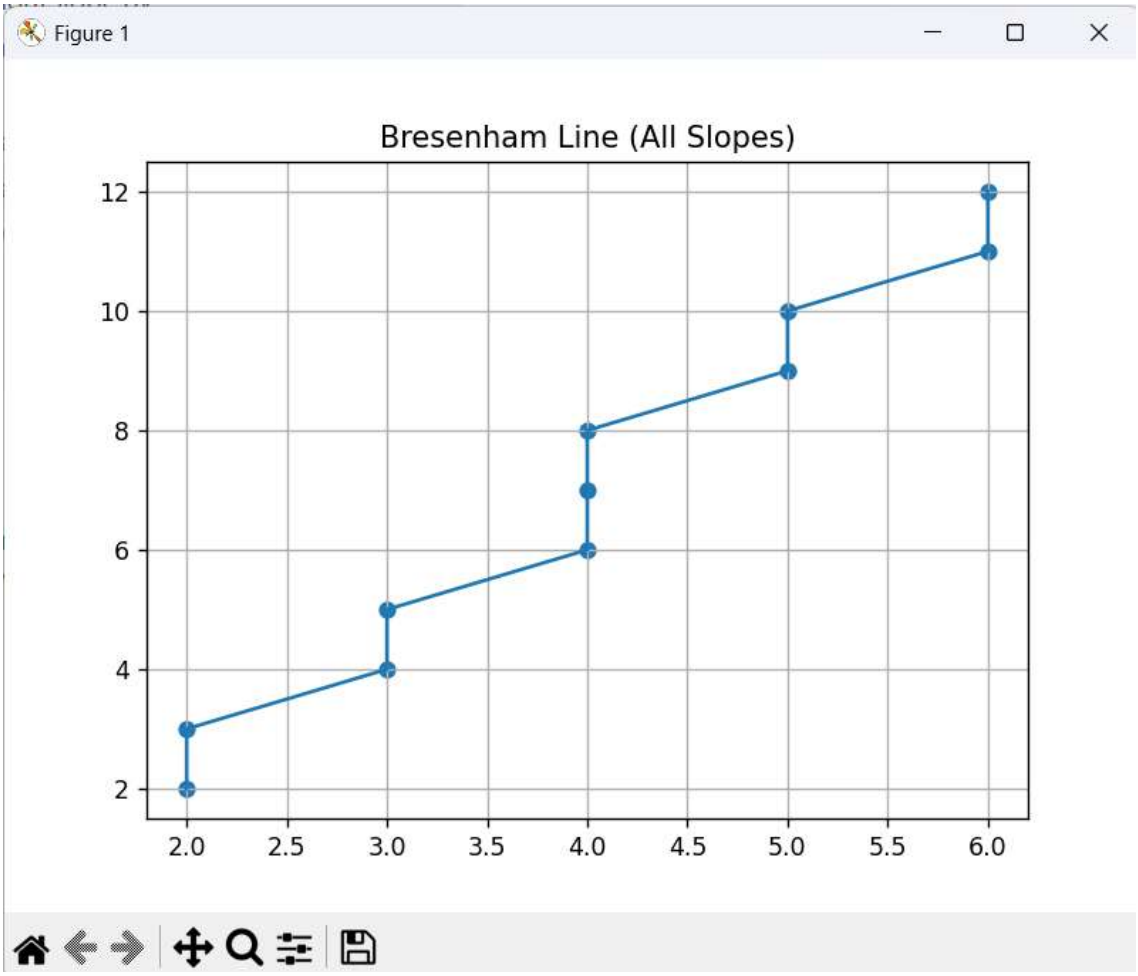


Figure 1
Bresenham Line (All Slopes)
12
10
8
6
4
2
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0